Dr. **Sushobhan Sen**
AB6/329A, Indian Institute of Technology Gandhinagar, Palaj, Gandhinagar - 382355, Gujarat, India
sushobhan.sen@iitgn.ac.in ∘ labs.iitgn.ac.in/bel ∘ linkedin.com/in/sushobhansen

EDUCATION

University of Illinois at Urbana-Champaign (UIUC) Urbana, IL, USA
Doctor of Philosophy in Civil Engineering December, 2019
Thesis topic: Role of Pavements in Urban Energetics *GPA: 4.0/4.0*
Committee members: Dr. Jeffery Roesler (UIUC, Chair), Dr. Imad Al Qadi (UIUC), Dr. Arif Masud (UIUC), Dr. Jeremy Gregory (MIT), and Dr. John Harvey (UC Davis)

University of Illinois at Urbana-Champaign (UIUC) Urbana, IL, USA
Master of Science in Civil Engineering August, 2015
Thesis Topic: Impact of Concrete Pavements on the Urban Heat Island *GPA: 4.0/4.0*
Thesis adviser: Dr. Jeffery Roesler (UIUC)

Indian Institute of Technology Roorkee Roorkee, India
Bachelor of Technology May, 2013
Major: Civil Engineering *GPA: 9.5/10.0*
Awarded the Institute Silver Medal

WORK EXPERIENCE

Indian Institute of Technology Gandhinagar
Assistant Professor December, 2022 - Present
Assistant professor of civil engineering and PI at the Built Environment Lab. Started a new masters and doctoral program in Transportation Engineering at IIT Gandhinagar.

Associate Dean, Campus and Space Management September, 2025 - Present

University of Pittsburgh
Postdoctoral Associate January, 2020 - December, 2022
Conducted research in fracture mechanics, finite element modeling, bonded concrete overlays, urban heat islands, and viscoelastic modeling. Supervisors: Dr. Lev Khazanovich and Dr. Julie Vandenbossche.

ePAVE, LLC
Technical Intern June, 2018 - August, 2018
Prepared material specifications for a cool pavement coating, co-authored a white paper on cool pavements, and assisted in writing an NSF SBIR proposal.

TEACHING EXPERIENCE

Indian Institute of Technology Gandhinagar

Instructor

Finite Element Method - Primary Instructor - Spring 2023-25
Pavement Materials and Design - Primary Instructor - Fall 2023-24
Traffic and Roadway Engineering - Primary Instructor - Spring 2024-25
Transportation Engineering - Primary Instructor - Fall 2024
Foundation Programme - Co-Instructor - Fall 2023-25

University of Illinois at Urbana-Champaign

CEE Transportation Instructional Fellow

Geometric Design of Roads - Primary Instructor - Spring 2017, Spring 2019

Graduate Teaching Assistant

Geometric Design of Roads - Teaching Assistant - Spring 2016, Spring 2018

Workshop Instructor

Introduction to Python programming - Primary Instructor - April 2019, October 2019

STUDENTS GUIDED

Postdocs

1. Dr Sumit Nandi, June, 2023 - Present
2. Dr Shubham Kumar Verma, November, 2023 - May, 2025

Doctoral students

1. Chirag Nagar, *Improving roadside urban microclimate using green and blue infrastructure*, December, 2027 (expected)
2. Ankita Singh, *AI/ML applications for rigid pavement analysis*, December, 2027 (expected)
3. Pratiksha, *An advanced numerical method for modeling transverse cracks in rigid pavements*, August, 2029 (expected)
4. Nydia Takhellambam, *Environmental stressors impacting last-mile connectivity for select metro stations in Ahmedabad*, May, 2030 (expected)

Masters students

1. Avinash Kumar Jha, *Improved Estimates of Temperature Distributions for Design of Jointed Plain Concrete Pavements in India and Nepal*, May, 2024 (winner of the **Best Paper Award** in the category of Pavements and Materials, 7th Conference of the Transportation Research Group of India, Surat, December, 2023)
2. Bhabya Thakur, *Analysis of jointed plain concrete pavements under heterogeneous traffic*, January, 2025

3. Anushka Khachi, *Representative Weather Datasets for Long-Term Temperature Predictions in Rigid Pavements*, May, 2025
4. Faizy Khan, *Effects of shade on roads in urban areas*, June, 2025
5. Shubham Yadav, *Public Transportation-Based Evacuation Planning for the Statue of Unity using Microscale Traffic Modelling*, May, 2025
6. Rubayet Jahan, *Development of Homogenized Interlayer Theory (HIT) for Geosynthetic Reinforced Flexible Pavements*, June, 2025 (co-supervised with Prof. Amit Prashant)
7. Mehnaaz Fatima, *Computational assessment of the efficacy of smog towers in cleaning ambient air*, June, 2025 (co-supervised with Prof. Sameer Patel)
8. Aashish Timilsina, *Impact of climate change-induced temperature changes on the performance of flexible pavements on Indian highways*, May, 2026
9. Md Asad Imam Khan, *Insights on paddy farmers in rural Bihar*, May, 2026
10. Khundongbam Preety Devi, *Structural response and failure patterns in geometrically varying concrete block pavements*, May, 2026
11. Keshar Lahare, *Thermal performance and building-scale energy simulation of textile-waste cement blocks and panels*, May, 2026
12. Neelam Shukla, *Road accident reporting using LLMs*, May, 2026
13. Saswati Das Choudhury, *Subgrade rutting prediction in geosynthetic reinforced flexible pavements using cyclic non-linear layered elastic theory*, June, 2026 (expected) (co-supervised with Prof. Amit Prashant)
14. Md Riaz Ali Mamud, *Layered viscoelastic theory for the analysis of flexible pavements*, August, 2026 (expected)
15. Anal Chaudhary, *An agentic AI platform for performing FE simulations on rigid pavement slabs*, May, 2027 (expected)
16. Abhirohan Kashyap, *Numerical modeling of dowel bar performance using peridynamics*, May, 2027 (expected)
17. Shrishti Raj, *Evacuation modeling of a student cafeteria using the social force model*, May, 2027 (expected)

Bachelors students

1. Sourabh Prajapati, *Development of FEM surrogate models for isotropic and orthotropic rigid pavement slabs*, Summer intern, 2025
2. Karthik Chandra Srinayak, *Drag coefficient of varying urban morphologies*, Summer intern, 2025
3. Shiv Patel, *Time-series AI models for thermal behaviour of rigid pavements*, Summer intern, 2025

4. Abhitej Singh Bhullar, *AI models for dynamical downscaling of climate data in India*, Summer intern, 2025
5. Om Gupta and Nirman Jaiswal, *NucleoSite: A GIS platform for planning for nuclear reactors in India*, Summer intern, 2024
6. Ashwini Ramanuj, *Characterizing urban morphology of Indian cities using Machine Learning*, Summer intern, 2023 (winner of the **Bhalodia-Khetan summer research excellence award**)
7. Abhinav Kumar, *Flexible pavement modelling using machine learning*, Summer intern, 2023

PROJECTS

Sponsored research

1. An advanced numerical technique for predicting transverse cracking in rigid pavement slabs (2025), ANRF-ECRG, Role: PI, Duration: 3 years, Amount: ₹34 lakhs
2. Development and Evaluation of Textile Waste-Cement Composite Building Material (2025), Truetzschler-IITGN Research Initiative, Role: Co-PI, Duration: 1 year, Amount: ₹12.4 lakhs
3. IndoULI: An Urban Livability Index for sustainable environmental monitoring of Indian cities (2024), Higher Education Financing Agency (HEFA), Role: Co-PI, Duration: 2 years, Amount: ₹42.0 lakhs
4. Roadside green and blue infrastructure for urban heat island mitigation (2023), IIT Gandhinagar, Role: PI, Duration: 3 years, Amount: ₹19.0 lakhs

Consultancy

1. Comprehensive Risk Assessment and Emergency Management Plan for Ekta Nagar (2026), Gujarat Institute of Disaster Management, Role: Co-PI, Amount: ₹60 lakhs
2. Verification of some bituminous material tests (2026), Nexxus Petro Industries Ltd, Role: PI, Amount: ₹4.05 lakhs
3. Testing of 6 nos PQC cores from Air Force Station Deesa (2026), Indian Air Force, Role: PI, Amount: Rs 46,000
4. Third-party quality inspection and technical assessment for Dholera International Airport, Gujarat (2025), Role: Co-PI, Amount (to Co-PI only): ₹15 lakhs
5. Risk analysis and protection of GMRC metro piers near railway lines in Ahmedabad, Gujarat (2025), Metro Rail Corporation Ltd, Role: PI, Amount: ₹14.75 lakhs
6. PCR analysis for HDD of Taxiways B, G, and H (2025), Ahmedabad International Airport Ltd, Role: PI, Amount: ₹2.5 lakhs
7. Testing of 24 Nos concrete cores from a section of NH-8E (2024), NHAI, Role: PI, Amount: ₹2.0 lakhs

8. Stopping seepage in subways in Gondal city area, district Rajkot (2024), Indian Railways, Role: Co-PI, Duration: 2 months, Amount: ₹11.8 lakhs
9. Verification of certain bituminous concrete mix designs (2024), RKC Infrabuilt Pvt Ltd, Role: PI, Duration: 10 days, Amount: ₹2.3 lakhs
10. Study of performance efficacy measurement of mobile quality control van deployed by Ministry of Road Transport and Highways (2024), MoRTH, Role: Co-PI, Duration: 6 months, Amount: ₹44.2 lakhs
11. Assessment of stretches of NH-47 from KM 60.000 to KM 105.000 in Gujarat (2023), NHAI, Role: PI, Duration: 2 months, Amount: ₹17.7 lakhs

REPORTS AND PUBLICATIONS

Peer-Reviewed Journal Articles

1. Khachi, A., and **Sen, S.** (2026) *Climate representativeness of a typical meteorological year for long-term temperature predictions in rigid pavements.* ASCE Journal of Transportation Engineering, Part B: Pavements, Vol. 152, No. 3, pp. 04026034, doi: 10.1061/JPEODX.PVENG-1929
2. Borah, A., **Sen, S.**, and Bhatia, U. (2026) *Heat-stress reduction through targeted green infrastructure using computational urban climate twins.* Climatic Change, Vol. 179, No. 6, doi: 10.1007/s10584-025-04091-3
3. Dave, R., **Sen, S.**, Maji, A., and Bhatia, U. (2025) *Local damages drive network-wide costs in compound flood-prone coastal city.* npj Urban Sustainability, Vol. 5, No. 102, doi: 10.1038/s42949-025-00288-8
4. Verma, S.K., Nagar, C., and **Sen, S.** (2025) *Improving urban sustainability through microscale configuration of green infrastructure.* Mitigation and Adaptation Strategies for Global Change, Vol. 30, No. 66, doi: 10.1007/s11027-025-10253-6
5. Khan, F., and **Sen, S.** (2025) *Thermal performance of jointed plain concrete pavements in urban areas considering shade from buildings.* Discover Civil Engineering, Vol. 2, No. 179, doi: 10.1007/s44290-025-00335-4
6. Nandi, S., Nagar, C., **Sen, S.**, Bansal, K., and Bhushan, V. (2025) *Spatio-temporal variation of microclimate in mixed-use urban forms.* Theoretical and Applied Climatology , Vol. 156, Art No. 490, doi: 10.1007/s00704-025-05734-2
7. Jha, A., and **Sen, S.** (2025) *Improved estimates of temperature distributions for design of jointed plain concrete pavements in India and Nepal.* International Journal of Pavement Engineering, Vol 26, No. 1, doi: 10.1080/10298436.2025.2503392
8. Jha, A., Thakur, B., and **Sen, S.** (2025) *Thermal Effects in Jointed Plain Concrete Pavements in Gujarat and Rajasthan: A Comparison with IRC:58.* Transportation Research Record, Available online **(awarded best paper in pavements and materials at CTRG 2025)**, doi: 10.1177/03611981251327210
9. Nagar, C., Verma, S.K., and **Sen, S.** (2025) *Synergistic deployment of green infrastructure and reflective pavements for mitigation of UHI within urban blocks.* Discover

Cities, Vol. 2, p. 7, doi: 10.1007/s44327-025-00046-6

10. Li, H., **Sen, S.**, and Khazanovich, L. (2024) *A scalable adaptive sampling approach for surrogate modeling of rigid pavements using machine learning.* Results in Engineering, Vol. 23, p. 102483, doi: 10.1016/j.rineng.2024.102483

11. Donnelly, C.A., **Sen, S.**, DeSantis, J.W., and Vandenbossche, J.M. (2024) *Prediction of effective equivalent linear temperature gradients in bonded concrete overlays of asphalt pavements.* Engineering Computations, Vol. 41 No. 2, pp. 468-485, doi: 10.1108/EC-04-2023-0161

12. Zhao, Y., **Sen, S.**, Susca, T., Iaria, J., Kubilay, A., Gunawardena, K., Zhou, X., Takane, Y., Park, Y., Wang, X., Rubin, A., Fan, Y., Yuan, C., Bardhan, R., Derome, D., Ürge-Vorsatz, D., Carmeliet, J. (2023) *Beating urban heat: Multimeasure-centric solution sets and a complementary framework for decision-making.* Renewable and Sustainable Energy Reviews, Vol. 186, p. 113668, doi: 10.1016/j.rser.2023.113668

13. Donnelly, C.A., **Sen, S.**, and Vandenbossche, J.M. (2023) *Fatigue Damage Prediction for Superload Vehicles in Pennsylvania on Jointed Plain Concrete Pavements.* Journal of Transportation Engineering, Part B: Pavements, Vol. 149, No. 4, p. 04023029, doi: 10.1061/JPEODX.PVENG-1334

14. Donnelly, C.A., **Sen, S.**, and Vandenbossche, J.M. (2023) *Reduction of critical positive temperature gradients in jointed plain concrete pavements.* International Journal of Pavement Engineering, Vol. 24, No. 1, doi: 10.1080/10298436.2023.2197645

15. **Sen, S.**, and Roesler, J. (2023) *D-SPARC: Rapid field albedo measurement.* Climate, Vol. 11, No. 3, p. 64, doi: 10.3390/cli11030064

16. **Sen, S.**, Li, H., and Khazanovich, L. (2022) *Effect of climate change and urban heat islands on the deterioration of concrete roads.* Results in Engineering, Vol. 16, p. 100736, doi: 10.1016/j.rineng.2022.100736

17. DeSantis, J.W., **Sen, S.**, and Vandenbossche, J.M. (2022) *Mechanistic-empirical model to predict transverse joint faulting of bonded concrete overlays of asphalt.* Road Materials and Pavement Design, Available online, doi: 10.1080/14680629.2022.2061371

18. Mainieri, J.G., **Sen, S.**, Roesler, J., and Al Qadi, I.L. (2022) *Albedo change mechanism of asphalt concrete surfaces.* Transportation Research Record, Vol. 2676, No. 7, pp. 763–772, doi: 10.1177/03611981221082567

19. **Sen, S.**, and Khazanovich, L. (2022) *Reconsidering the strength of concrete pavements.* International Journal of Pavement Engineering, Vol. 24, No. 2, p. 2020270 doi: 10.1080/10298436.2021.2020270

20. **Sen, S.**, and Khazanovich, L. (2021) *Limited application of reflective surfaces can mitigate urban heat pollution.* Nature Communications, Vol. 12, No. 3491, pp. 1-8, doi: 10.1038/s41467-021-23634-7

21. **Sen, S.**, and Khazanovich, L. (2021) *A self-contained element for modeling crack propagation in beams.* Engineering Fracture Mechanics, Vol. 242, pp. 107460, doi: 10.1016/j.engfracmech.2020.107460

22. **Sen, S.**, Fernandéz, J.P.R.M., and Roesler, J. (2020) *Reflective Parking Lots for Microscale Urban Heat Island Mitigation.* Transportation Research Record: Journal of the Transportation Research Board, Vol. 2674, No. 8, pp. 663-671, doi: 10.1177/0361198120919401
23. **Sen, S.**, and Roesler, J. (2020) *Wind direction and cool surface strategies on microscale urban heat island.* Urban Climate, Vol. 31, p. 100548, doi: 10.1016/j.uclim.2019.100548
24. **Sen, S.**, Roesler, J., Ruddell, B., and Middel, A. (2019) *Cool pavement strategies for urban heat island mitigation in suburban Phoenix, Arizona.* Sustainability, Vol. 11, No. 16, 4452, doi: 10.3390/su11164452
25. **Sen, S.**, and Roesler, J. (2019) *Thermal and optical characterization of asphalt field cores for microscale urban heat island analysis.* Construction and Building Materials, Vol. 217, pp. 600-611, doi: 10.1016/j.conbuildmat.2019.05.091
26. Baral, A., **Sen, S.**, and Roesler, J. (2018) *Use-phase assessment of photocatalytic cool pavements.* Journal of Cleaner Production, Vol. 190, pp. 722-728, doi: 10.1016/j.jclepro.2018.04.155
27. **Sen, S.**, Roesler, J., and King, D. (2019) *Albedo estimation of finite-sized concrete specimens.* Journal of Testing and Evaluation, Vol. 47, No. 2, pp. -, doi: 10.1520/JTE20170059
28. **Sen, S.**, and Roesler, J. (2017) *Microscale heat island characterization of rigid pavements.* Transportation Research Record: Journal of the Transportation Research Board, No. 2639, pp. 73-83, doi: 10.3141/2639-10
29. **Sen, S.**, and Roesler, J. (2016) *Aging albedo model for asphalt pavements.* Journal of Cleaner Production, Vol. 117, pp. 169-175, doi: 10.1016/j.jclepro.2016.01.019
30. **Sen, S.**, and Roesler, J. (2016) *Contextual heat island assessment for pavement preservation.* International Journal of Pavement Engineering, Vol. 19, No. 10, pp. 865-873, doi: 10.1080/10298436.2016.1213842

Peer-Reviewed Conference Papers

1. Khachi, A., and **Sen, S.** (2025). "Weather" art thy climate? Climate representativeness in rigid pavement design. Proceedings of the 8th ASCP Concrete Pavements Conference, Penrith, NSW, Australia, doi: 10.31224/6816
2. Yadav, S., and **Sen, S.** (2025). Evacuation Routing Using Public Transportation Around the Statue of Unity. In: Saha, S., Biswas, S. (eds) Innovations for Sustainable and Resilient Infrastructure (ICSRI 2024), ISBN 978-3-031-91976-3, pp. 268–278, doi: 10.1007/978-3-031-91976-3_26
3. Verma, S.K., and **Sen, S.** (2025). Leveraging Green Infrastructure to Mitigate Urban Heat and Energy Consumption: A Microscale Perspective. In: Saha, S., Biswas, S. (eds) Innovations for Sustainable and Resilient Infrastructure (ICSRI 2024), ISBN 978-3-031-91976-3, pp. 257–267, doi: 10.1007/978-3-031-91976-3_25

4. Nandi, S., Bansal, K., Bhushan, V., and **Sen, S.** (2025). *Sky View Factor Analysis for a Compact Self-Contained Urban University Campus: A Case Study of IIT Gandhinagar.* American Society of Civil Engineers (ASCE) India Section Northern Region Conference on Challenges and Innovations in Sustainable Smart Cities, Chandigarh, India **(awarded best paper in the category of urban planning and development)**, doi: 10.22541/essoar.174708024.46602429/v1
5. Kennebeck, K., Khazanovich, L., **Sen, S.**, Smetana, M., and Mu, F. (2024). *Developing a Next-Generation Concrete Pavement Analysis Tool: PITTSLAB.* 13th International Conference on Concrete Pavements, Minneapolis, MN, USA **(awarded the ACPA Robert G. Packard Award for Design Innovations in Concrete Pavements)**, doi: 10.33593/txxp6c32
6. Roesler, J., and **Sen, S.** (2024). *Concrete thermal and optical property sensitivities to mixture proportions.* 13th International Conference on Concrete Pavements, Minneapolis, MN, USA, doi: 10.33593/9qjt8b62
7. Li, H., **Sen, S.**, and Khazanovich, L. (2024). *Artificial Neural Networks for Mechanistic-Empirical Concrete Pavement Design: An Efficient Development Approach.* 13th International Conference on Concrete Pavements, Minneapolis, MN, USA, doi: 10.33593/517z0c86
8. Stolte, S., Pierce, L., Weitzel, N., Medina, J., **Sen, S.**, Scott, G., and Roesler, J. (2022) *Field and laboratory evaluation of in-service bonded concrete overlays on asphalt.* 101st Annual Meeting of the Transportation Research Board, Washington, DC.
9. **Sen, S.**, Roesler, J., Ruddell, B., and Middel, A. (2021) *Cool pavements for sustainable urban development.* 12th International Conference on Concrete Pavements, virtual conference, ISBN 978-0-578-33418-9, pp. 623-627, doi: 10.33593/xx1hzrq3
10. **Sen, S.**, and Roesler, J. (2021) *Heat island impact of chip seals.* Airfield and Highway Pavements 2021: Pavement Materials and Sustainability, virtual conference, pp. 320-331, doi: 10.1061/9780784483510.029
11. **Sen, S.**, and Roesler, J. (2021) *Human-centered geometric design of roads using an autonomous vehicle problem.* American Society of Engineering Education Middle Atlantic Section Spring 2021 Conference, Villanova, PA, link: peer.asee.org/36302
12. **Sen, S.**, and Roesler, J. (2020) *Rapid ground-based measurement of pavement albedo.* Pavement, Roadway, and Bridge Life Cycle Assessment 2020, Sacramento, CA, ISBN 9781003092278, pp. 533-540, doi: 10.1201/9781003092278-55
13. **Sen, S.**, and Roesler, J. (2019) *Coupled pavement-urban canyon model for cool pavements assessment.* Airfield and Highway Pavements 2019: Innovation and Sustainability in Highway and Airfield Pavement Technology, Chicago, IL, USA, pp. 207-215, doi: 10.1061/9780784482476.022
14. Baral, A., **Sen, S.**, and Roesler, J. (2018) *Environmental design concept for multifunctional concrete overlays.* 9th International DUT-Workshop on Research and Innovations for Design of Sustainable and Durable Concrete Pavements, Potsdam, Germany

15. **Sen, S.**, and Roesler, J. (2017) *Pavement geometry in microscale urban heat islands.* Conference of the Transportation Association of Canada, St. John's, NL, Canada (link)
16. **Sen, S.**, Baral, A., and Roesler, J. (2017) *Use-phase sustainability through preservation.* 10th International Conference on Road and Airfield Pavement Technology, Hong Kong
17. **Sen, S.**, and Roesler, J. (2017) *An uncoupled pavement-urban canyon model for heat islands.* International Symposium on Pavement Life Cycle Assessment, Champaign, IL, ISBN 978-1-315-15932-4, pp. 111-120, doi: 10.1201/9781315159324-13
18. **Sen, S.**, and Roesler, J. (2016) *Albedo as an engineering property of concrete pavements.* 11th International Conference on Concrete Pavements, San Antonio, TX, ISBN 978-0-9860291-2-7, pp. 59-71 **(awarded best paper by a young author)**, doi: 10.33593/iccp.v11i1.270
19. King, D., Roesler, J., and **Sen, S.** (2016) *Emissions reducing benefits of multifunctional photocatalytic concrete inlays.* 11th International Conference on Concrete Pavements, San Antonio, TX, ISBN 978-0-9860291-2-7, pp. 72-81, doi: 10.33593/iccp.v11i1.271
20. **Sen, S.**, King, D., and Roesler, J. (2015) *Structural and environmental benefits of concrete inlays for pavement preservation.* Airfield and Highway Pavements 2015, Miami, FL, pp. 697-707, doi: 10.1061/9780784479216.062
21. **Sen, S.**, and Roesler, J. (2014) *Assessment of concrete pavement structure on urban heat island.* International Symposium on Pavement Life Cycle Assessment, Davis, CA, pp. 191-200 (link)

Book Chapters

1. **Sen, S.** (2025) *Geoengineering Cities with Reflective and Pervious Surfaces.* Geoengineering and Climate Change: Methods, Risks, and Governance, Martin Beech (Editor), Wiley-Scrivener, ISBN 978-1-394-20438-0, doi: 10.1002/9781394204847.ch14

Theses

1. **Sen, S.** (2019) *Role of pavements in urban energetics.* University of Illinois at Urbana-Champaign, Urbana, IL. Doctoral dissertation
2. **Sen, S.** (2015) *Impact of concrete pavements on the Urban Heat Island.* University of Illinois at Urbana-Champaign, Urbana, IL. Master's Thesis

Reports

1. Pierce, L.M., Stolte, S.E., Weitzel, N., Medina, J., Van Dam, T., Senn, K., Roesler, J., Scott, G.M., **Sen, S.**, Jadallah, O.A., Maser, K., Carmichael, A., Smith, K., and Smith, K. (2022) *NCHRP Research Report 1007: Evaluation of Bonded Concrete Overlays on Asphalt Pavements*, National Cooperative Highway Research Program (NCHRP)

Project 01-61, National Academies of Sciences, Engineering, and Medicine (NASEM), Washington, DC, doi: 10.17226/26760

2. Vendenbossche, J.M., Donnelly, C.A., Buettner, N., **Sen, S.**, and Brody, Z.A. (2022) *Effect of Superloads on Pavement Life.* Pennsylvania Department of Transportation Report No. FHWA-PA-2022-001-PITT WO 019, University of Pittsburgh, Pittsburgh, PA (link)
3. Khazanovich, L., Vandenbossche, J.M., Salles, L., **Sen, S.**, Donnelly, C.A., and Kosar, K. (2021) *Faulting model improvements for MEPDG.* Pennsylvania Department of Transportation Report No. FHWA-PA-2021-005-PITT WO 001, University of Pittsburgh, Pittsburgh, PA (link)
4. Zhou, Q., Okte, E., **Sen, S.**, Ozer, H., Al-Qadi, I. L., Roesler, J. R., and Chatti, K. (2019) *Development of a Life-cycle Assessment Tool for Pavement Preservation and Maintenance on Flexible and Rigid Pavement Vol I.* Illinois Center for Transportation Series No. ICT-19-010, Illinois Center for Transportation, UIUC, Rantoul, IL (link)
5. Zhou, Q., Okte, E., **Sen, S.**, Rajaei, S., Ozer, H., Al-Qadi, I. L., Roesler, J. R., and Chatti, K. (2019) *Development of a Life-cycle Assessment Tool for Pavement Preservation and Maintenance on Flexible and Rigid Pavement Vol II.* Illinois Center for Transportation Series No. ICT-19-011, Illinois Center for Transportation, UIUC, Rantoul, IL (link)
6. **Sen, S.**, and Moradian, K. (2018) *Better pavements make better communities: the promise of cool pavements.* White paper for ePAVE, LLC, Los Angeles, CA (link)

Intellectual Property

1. **Sen, S.**, Li, H., and Khazanovich, L. *A scalable adaptive sampling method for surrogate modeling using active machine learning.* Provisional US patent filed (2022).
2. Khazanovich, L. and **Sen, S.** *A quasi-elastic finite element method program for analysis of the effect of concrete solidification on the long-term behavior of beams-on-grade.* Disclosure filed (2021).
3. **Sen, S.**, Roesler, J., and Dahal, S. *D-SPARC: A Discrete Spectral Reflectometer for Rapid Albedo Measurement.* Disclosure filed (2020).

HONORS AND AWARDS

13th *International Conference on Concrete Pavements*
ACPA Robert G. Packard Award for Design Innovations in Concrete Pavements
September, 2024

Indian Institute of Technology Gandhinagar
Excellence in Research Fellowship December, 2022

Illinois Center for Transportation
PhD Student Symposium Award October, 2017
Awarded second place among a cohort of 50 top international PhD students in pavement engineering.

11th *International Conference on Concrete Pavements*
Bengt F. Friberg Award for Best Paper by a Young Author August, 2016

University of Illinois at Urbana-Champaign
Mavis Future Faculty Fellowship August, 2017 - July, 2018
Awarded a fellowship among engineering PhD students to be trained for a career in academia.

CEE Transportation Instructional Fellowship January, 2017 & 2019
Awarded a fellowship to teach a senior design course in the Department of Civil and Environmental Engineering.

Teacher Scholar Certificate May, 2018
Earned a certificate in pedagogical scholarship by designing teaching materials, engaging with the literature, and gaining extensive teaching experience.

Graduate Teacher Certificate May, 2018
Earned a certificate in teaching based on teaching experience and participation in teacher training activities.

Ravindar K and Kavita Kinra Fellowship August, 2013 - July, 2014
Earned a competitive private fellowship to engage in research for 1 year as an MS student.

Indian Institute of Technology Roorkee
Institute Silver Medal and Other Honors October, 2013
Earned several awards for finishing at the top of the Civil Engineering class of 2013.

German Academic Exchange Service (DAAD)
Working Internships in Science and Engineering (WISE) Scholarship May, 2012
Awarded a scholarship to engage in summer research at Technical University Munich (TUM).

INVITED TALKS

1. *A hundred years of rigid pavement analysis.* Invited talk to the Department of Civil Engineering, IIT Madras (2026)
2. *Mechanistic-Empirical Pavement Design.* Invited talk to the Faculty Development Programme, Department of Civil Engineering, Darshan University, Rajkot (2024)

3. *Advanced topics in finite element modeling.* Invited talk to Introduction to the Finite Element Method class (CEE 2333), Department of Civil and Environmental Engineering, University of Pittsburgh (2022)
4. *Climate change adaptation of pavements at local scale.* Invited talk to the Department of Civil Engineering, Indian Institute of Technology Gandhinagar (2021)
5. *Adapting neighborhood pavements for climate change.* Invited talk to the Department of Civil Engineering, Indian Institute of Technology Kanpur (2021)
6. *Pavement materials in the context of the urban heat island.* Invited talk at the short term course on Advances in Pavement Engineering, Indian Institute of Technology Bhubaneswar (2021)
7. *Microscale urban heat islands: cooling your neighborhood.* Invited talk to the Department of Civil and Environmental Engineering, Carnegie Mellon University (2020)
8. *Phase change materials for urban heat island mitigation.* Invited talk at webinar organized by the Transportation Research Board Committee on Sustainable and Resilient Pavements (2020)

MEDIA COVERAGE

1. *Large parks, reflective spaces can beat urban heat: Study.* Times of India Ahmedabad Edition, August 8, 2025. (link)
2. *Betongveier rammes hardt av klimaendringene.* Byggeindustrien, December 27, 2022. (link) [In Norwegian]
3. *Cambio climático deteriora el pavimento de concreto.* SciDevNet, December 12, 2022. (link) [In Spanish]
4. *Concrete roads could deteriorate faster because of climate change.* Pittsburgh Union Progress, November 22, 2022. (link)
5. *Estratégias de resfriamento urbano usando superfícies reflexivas.* EcoDebate, December 12, 2021. (link) [In Portuguese]
6. *Creating cooler cities.* EurekAlert! American Association for the Advancement of Science, June 21, 2021. (link)
7. *Research Update: A Few Changes to Mitigate UHI Effect and Cool Cities.* International Society for Concrete Pavements, June 15, 2022. (link)
8. *Civil engineers examine urban cooling strategies using reflective surfaces.* Science Daily, June 21, 2021. (link)
9. *Engineers examine urban cooling strategies using reflective surfaces.* Phys.org, June 22, 2021. (link)
10. *UIUC grad has the remedy for surging urban temps.* Illinois Center for Transportation, February 3, 2020. (link)
11. *Urban heat islands studied within a pavement LCA framework.* Illinois Center for

Transportation, May 1, 2018. (link)

12. *Student spotlight: Sushobhan Sen.* Illinois Center for Transportation, February 1, 2018. (link)
13. *Wired In: Sushobhan Sen.* The News-Gazette, December 23, 2016. (link)

LEADERSHIP AND SERVICE

Transportation Research Board

TRB Sustainable and Resilient Pavements Subcommittee January 2014 - Present
Presented in annual meetings and sessions as a friend of the committee, helped organize a widely-attended webinar on urban heat islands and pavements, and co-authored a technical circular on cool pavements.

International Society for Concrete Pavements

Secretary-Treasurer August, 2025 - Present
Board Member June, 2023 - Present
Co-Director of Communications May, 2021 - August, 2025
Webmaster and student member July, 2015 - December, 2019

12th International Conference on Concrete Pavements

Organizing Team Member June - October, 2021
Organized The Student Pitch networking event and maintained the conference website.

University of Pittsburgh Postdoctoral Association (UPPDA)

University Senate Bylaws and Procedures Committee September, 2020 - May, 2022
Postdoctoral representative to the committee

UPPDA Gender and Racial Equity Sub-committee September, 2020 - August, 2021
Coordinator for the affinity group for postdocs from under-represented backgrounds

University of Illinois at Urbana-Champaign

ITE UIUC Student Chapter August, 2015 - August, 2017
Secretary of the Institute of Transportation Engineers (ITE) Student Chapter.

IGSA at UIUC August, 2013 - August, 2019
President of the Indian Graduate Students Association (IGSA) at UIUC from August, 2014 - August, 2015 and webmaster from August, 2015 - 2018